

Dongfang Zhao, Ph.D.

Homepage: hpdic.github.io

Professional Preparation

Katholieke Universiteit Leuven, Belgium	Statistics, Diploma	2005
Emory University, Atlanta, GA	Computer Science, MS	2008
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
University of Washington, Seattle	Computer Science, PostDoc	2017

Employment

2023–present	Tenure-Track Assistant Professor, University of Washington
2020–2022	Lead Developer, Moka Blox LLC, Davis, CA
2018–2020	Visiting Professor, University of California, Davis
2018–2022	Assistant Professor, University of Nevada, Reno
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer, Epic Systems Corporation, Madison, WI
2006–2007	Research Intern, Centers for Disease Control and Prevention, Atlanta, GA
2003–2004	Research Intern, Interuniversity Microelectronics Centre, Belgium

Selected Awards

2022	Federal Research Grant, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000

Selected Services

2021–present	Editorial Board	IEEE Transactions on Distributed & Parallel Systems
2018–present	Grant Review Panels	NSF CISE IIS, NSF CISE OAC, DOE ASCR
2023	Search Committee	One Associate Professor of Artificial Intelligence
2021	Search Committee	Two Assistant Professors of Cybersecurity
2019	Organizing Committee	ACM International Middleware Conference, UC Davis

Selected Publications

(**Dongfang Zhao** is the student's *primary advisor, [†]thesis committee member, or [‡]postdoc co-author)

- [1] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data, SIGMOD 2023.
- [2] *Abdullah Al-Mamun, **Dongfang Zhao**. ECHAIN: Securing Electronic Device Provenance through Privacy-Preserving Consortium Blockchains. The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023.
- [3] *Muhammad Jahanzeb Khan, *Olamide Timothy Tawose, Zoey Rui Hu, **Dongfang Zhao**. Exploring the Efficacy of Data-Decoupled Federated Learning for Image Classification and Medical Imaging Analysis. International Workshop on Federated Learning for Distributed Data Mining, co-located with the 29th ACM SIGKDD Conference (KDD), 2023.
- [4] **Dongfang Zhao**. Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. In *Handbook on Blockchain*, pp. 157-193. Cham: Springer International Publishing, 2022.
- [5] *Lesi Wang, **Dongfang Zhao**. Practical Federated Learning Infrastructure for Privacy-Preserving Scientific Computing. The 3rd Workshop on Artificial Intelligence and Machine Learning for Scientific Applications (AI4S), collocated with Supercomputing Conference (SC), 2022.
- [6] *Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data. 8(5): 1259-1272 (2022).
- [7] *Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [8] *Olamide Timothy Tawose, *Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance
- [9] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [10] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [11] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [12] *Abdullah Al-Mamun, Jun Dai, Xiaohua Xu, Mohammad Sadoghi, Haoting Shen, **Dongfang Zhao**. Poster DEAN: A Blockchain-Inspired Consensus Protocol Enabling Trustworthy Edge Computing. Network and Distributed System Security Symposium (NDSS). 2020.
- [13] *Abdullah Al-Mamun, Jialin Liu, Tonglin Li, Quincey Koziol, Zhongyi Zhai, Junyan Qian, Haoting Shen, **Dongfang Zhao**. Poster Reflector: a fine-grained I/O tracker for HPC systems. The 25th ACM Symposium on Principles and Practice of Parallel Programming (PPoPP), 2020.

- [14] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. *IEEE Transactions on Cloud Computing*. 8(2): 635-646 (2020).
- [15] *Xinying Wang, *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. *The 12th International Conference on Cloud Computing (CLOUD)*, 2019. Best paper award
- [16] †Heyang Qin, †Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. *The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC)*, 2019.
- [17] *Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. *The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD)*, 2018. Best student paper award
- [18] *Jiayao Wang, *Abdullah Al-Mamun, Tonglin Li, Linhua Jiang, **Dongfang Zhao**. Toward Performant and Energy-efficient Queries in Three-tier Wireless Sensor Networks. *Proceedings of the 47th International Conference on Parallel Processing (ICPP)*, 2018.
- [19] ‡Parmita Mehta, ‡Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. *Proceedings of the VLDB Endowment, VLDB 2017*.
- [20] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. *IEEE Transactions on Big Data*. 3(1): 107-117 (2017).
- [21] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, Ioan Raicu. Exploiting multi-cores for efficient interchange of large messages in distributed systems. *Concurrency and Computation: Practice and Experience*. 28(13): 3568-3585 (2016).
- [22] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, Ioan Raicu. Toward high-performance key-value stores through GPU encoding and locality-aware encoding. *Journal of Parallel and Distributed Computing*. 96: 27-37 (2016).
- [23] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. *IEEE Transactions on Parallel and Distributed Systems*. 27(6): 1824-1837 (2016).
- [24] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. *IEEE Transactions on Services Computing*. 9(1): 96-109 (2016).
- [25] **Dongfang Zhao**, Kent Burlingame, Corentin Debains, Pedro Alvarez-Tabio, Ioan Raicu. Towards high-performance and cost-effective distributed storage systems with information dispersal algorithms. *IEEE International Conference on Cluster Computing (CLUSTER)*, 2013.
- [26] **Dongfang Zhao**, Chen Shou, Tanu Malik, Ioan Raicu. Distributed data provenance for large-scale data-intensive computing. *International Conf. on Cluster Computing (CLUSTER)*, 2013.
- [27] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 31(1): 86-98 (2009).